ISSN 2319-5991 www.ijerst.com
Vol. 22, No. 2(1), 2026

NSRIT SmartBus Tracker Using Android with Real Time GPS Monitoring

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1.Abstract

The NSRIT SmartBus Tracker is an Android-based real-time bus tracking system designed to improve campus transportation efficiency and management. Unlike conventional systems that rely on dedicated GPS hardware, the proposed system uses a Virtual GPS Driver Module within a mobile application, where the driver's smartphone transmits real-time location data to Firebase Realtime Database. The system follows a cloud-based architecture to synchronize live bus data among students, drivers, and administrators with secure role-based authentication. Google Maps API enables real-time tracking and accurate Estimated Time of Arrival (ETA) prediction. It also includes an Admin Portal for managing users, monitoring activities, and controlling system operations. A digital fee payment module allows students to pay transportation fees securely through the app, reducing manual effort. Overall, the system reduces infrastructure costs, improves scalability, and provides an integrated solution for smart campus transportation management.

2.Keywords

Real-Time GPS Tracking, Android Application, Firebase Realtime Database, Google Maps API, Campus Transportation System, ETA Calculation, Cloud-Based Architecture, Role-Based Access Control.

3. Introduction

Transportation management in educational institutions is often handled through manual methods such as phone communication, notice boards, and fixed schedules, leading to delays, uncertainty, and lack of transparency. Students face difficulty in estimating bus arrival times, causing unnecessary waiting and safety concerns, while the absence of centralized monitoring limits administrative control and efficiency.

To address these issues, the NSRIT SmartBus Tracker proposes an Android-based real-time bus tracking system using modern mobile and cloud technologies. Instead of dedicated GPS hardware, it adopts a Virtual GPS Driver Model, where the driver's smartphone captures and transmits location data such as latitude, longitude, speed, and timestamp.

The data is sent to Firebase Realtime Database, enabling real-time synchronization for all users. Students can track buses and view ETA, while administrators monitor operations through a centralized system. The application also includes an Admin Portal for user management and a digital fee payment module for secure and convenient transactions.

Developed using Kotlin, Jetpack Compose, Firebase services, and Google Maps API, the system ensures real-time tracking, secure access, and efficient data handling. By eliminating hardware dependency, it reduces cost, improves scalability, and provides an integrated solution for smart campus transportation management.

4.System Architecture

4.1 Presentation Layer

The Presentation Layer is the front-end of the system, developed as an Android application using Kotlin and Jetpack Compose. It provides a user-friendly and responsive interface for students, drivers, and administrators with role-based dashboards. Students can view real-time bus location, routes, and ETA, while drivers can start or stop trips to enable GPS tracking. The Admin interface allows management of users, monitoring of bus activity, and system control through an Admin Portal. The layer also includes a digital fee payment interface for students, displaying payment status and transaction details. Overall, it ensures smooth interaction, real-time updates, and secure access to the system.

4.2 Application Layer

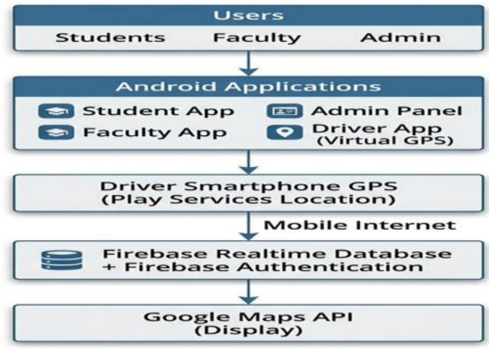
The Application Layer acts as the core logic of the system, handling authentication, real-time data processing, role-based access control, and ETA calculation. The driver's smartphone functions as a Virtual GPS transmitter using the Play Services Location API to capture location details such as latitude, longitude, speed, and timestamp.

This data is transmitted to Firebase Realtime Database and synchronized instantly across all users. Firebase Authentication ensures secure login and access for Student, Driver, and Admin roles. The ETA is calculated using real-time distance and speed data, providing accurate arrival predictions.

Additionally, this layer manages trip status (start/stop), user session handling, and real-time updates without manual refresh. It also supports admin functionalities such as monitoring user activity and handling system operations efficiently. Role-Based Access Control (RBAC) ensures data security, controlled access, and smooth system performance.

4.3 Data Layer

The Data Layer is responsible for storing, managing, and synchronizing all system data using Firebase Realtime Database. It maintains structured data for users, buses, routes, trip details, payment records, and real-time location updates. The driver's mobile GPS continuously sends location data to the cloud, which is instantly reflected across all connected users. This ensures accurate real-time tracking and seamless data flow. The layer also supports secure storage of user credentials, transaction details for fee payments, and system logs for monitoring purposes. Firebase provides high availability, automatic synchronization, and scalability, ensuring reliable performance and efficient data management across the system.



5. Modules Description

5.1 Student Module

The Student Module allows users to track bus locations in real time through an interactive map interface. Students can view bus routes, current location, and Estimated Time of Arrival (ETA), helping them plan their travel efficiently.

Additionally, this module includes a digital fee payment feature, enabling students to pay transportation fees securely through the application. Students can also view payment status and transaction history. The module ensures a simple, user-friendly interface with secure login and real-time updates.

5.2 Driver Module

The Driver Module is responsible for transmitting real-time bus location data. The driver uses the mobile application to start and stop trips, which activates the device's GPS and begins sending location details such as latitude, longitude, speed, and timestamp to the cloud.

This module eliminates the need for external GPS hardware by using the smartphone as a Virtual GPS transmitter. It ensures continuous and accurate location updates with minimal user interaction.

5.3 Admin Module

The Admin Module provides centralized control over the entire system through an Admin Portal. Administrators can manage user registrations, monitor login activities, and maintain records of students and drivers.

Admins can also monitor live bus movements, assign routes, and oversee system operations. In addition, the module allows tracking of fee payments, verifying transactions, and maintaining financial records. This module ensures efficient system management, data control, and overall operational transparency.

6. Methodology

6.1 GPS Data Acquisition

The system utilizes the built-in GPS module of the driver's Android smartphone to capture geographical coordinates, including latitude and longitude. Unlike traditional tracking systems, no dedicated GPS hardware device is installed in the bus. Instead, the driver's mobile device operates as a Virtual GPS transmitter. When the driver initiates a trip, the application activates the Google Play Services Location API to retrieve precise location updates at configured time and distance intervals. This approach ensures optimized battery consumption while maintaining high location accuracy. The captured data includes latitude, longitude, speed, and timestamp, forming the foundation for real-time tracking and ETA computation.

6.2 Data Transmission

Once the GPS coordinates are obtained, the location data is transmitted via mobile internet connectivity to the cloud backend. The system uses secure HTTPS protocols to ensure encrypted communication between the driver's mobile device and Firebase services. The location updates are pushed to the Firebase Realtime Database in structured JSON format. This cloud-based transmission mechanism ensures minimal latency, reliable delivery, and real time availability of tracking data even under moderate network conditions.

6.3 Real-Time Synchronization

Firebase Realtime Database enables instantaneous synchronization of data across all authenticated clients. The Android application implements Firebase event listeners that continuously monitor database changes.

Whenever the driver's device uploads updated location data, the changes are immediately reflected in the Student, Faculty, and Admin applications without requiring manual refresh. This ensures seamless communication between the mobile client and cloud infrastructure, maintaining accurate and up-to-date tracking information.

6.4 Map Visualization

The Google Maps API is integrated into the Android application to provide an interactive and dynamic visual interface for tracking buses. Live location coordinates retrieved from Firebase are plotted as dynamic markers on the map. As new location updates are received, the bus marker is updated smoothly to reflect real-time movement. Marker animation techniques are implemented to visually represent direction and motion, improving user clarity and navigation awareness.

6.5 ETA Calculation

The Estimated Time of Arrival (ETA) is calculated dynamically using the formula:

$$ETA = Distance/Speed$$

Where:

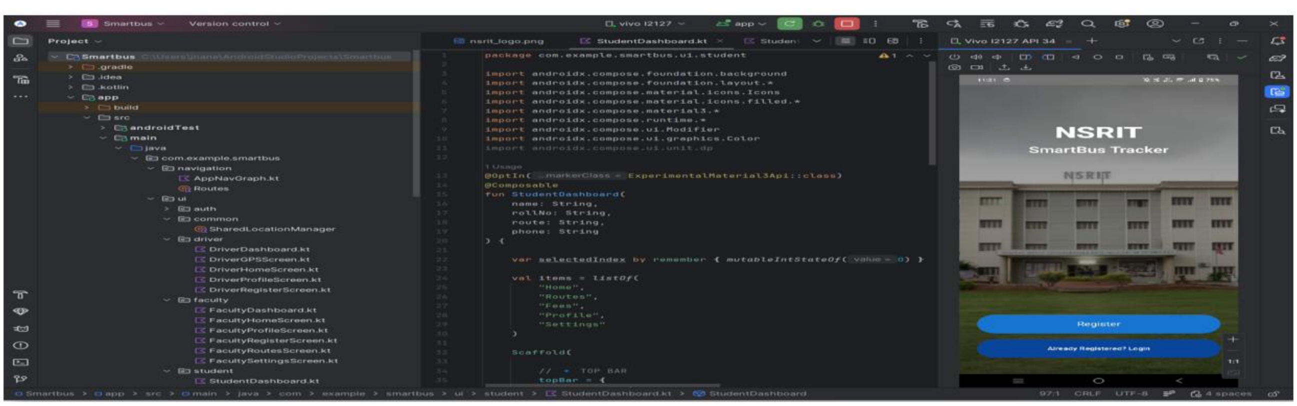
• Distance is calculated using the Haversine formula based on geographic coordinates.

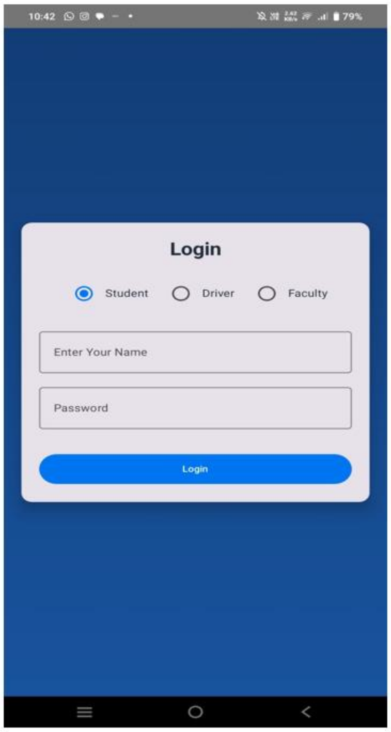
• Speed is derived from real-time GPS updates provided by the driver's mobile device.

This method ensures accurate and continuously updated arrival time predictions.

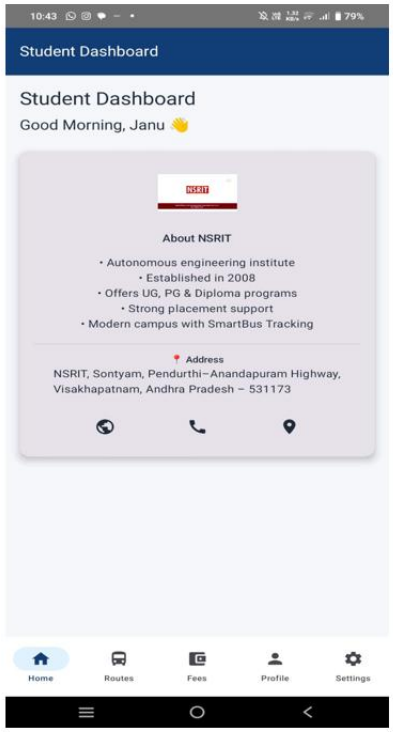
7. Results and Discussion

The system was tested in a simulated campus environment using real-time location updates from the driver's Android smartphone operating in Virtual GPS mode. The evaluation focused on synchronization speed, ETA accuracy, system responsiveness, secure role-based access, and payment functionality.

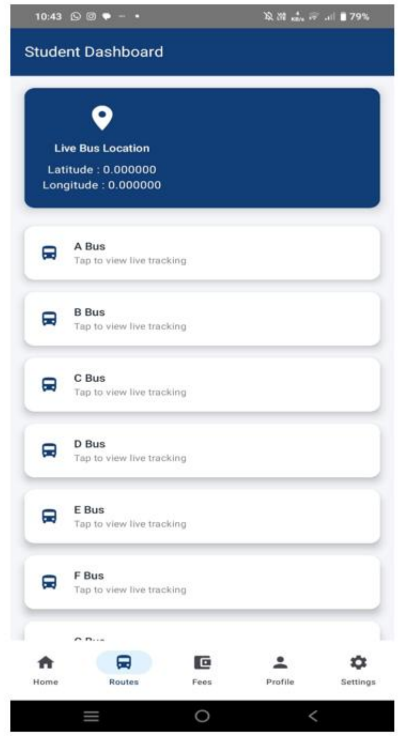




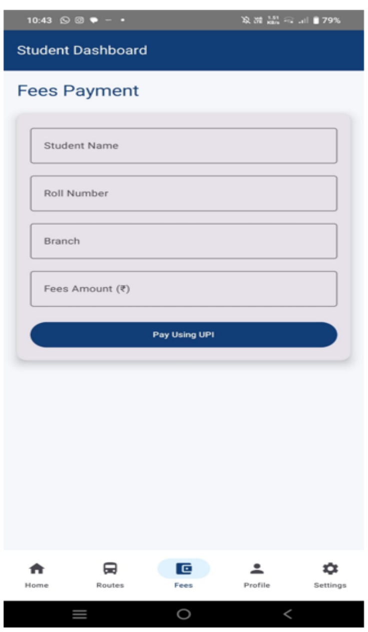
10:42
79%
Login
Student
Driver
Faculty
Enter Your Name
Password
Login



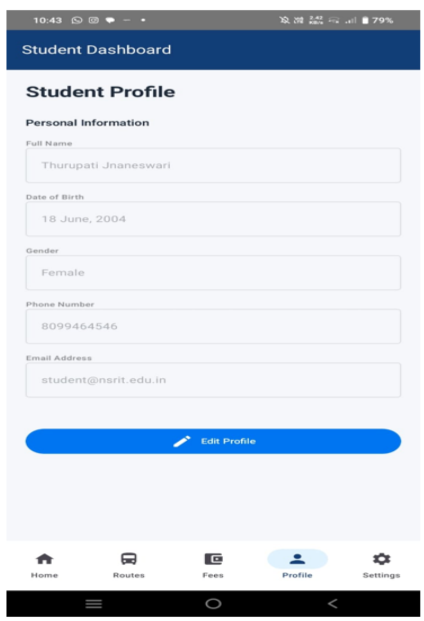
10:43
79%
Student Dashboard
Student Dashboard
Good Morning, Janu
NSRIT
About NSRIT
• Autonomous engineering institute
• Established in 2008
• Offers UG, PG & Diploma programs
• Strong placement support
• Modern campus with SmartBus Tracking
Address
NSRIT, Sontyam, Pendurthi–Anandapuram Highway,
Visakhapatnam, Andhra Pradesh – 531173
Home
Routes
Fees
Profile
Settings



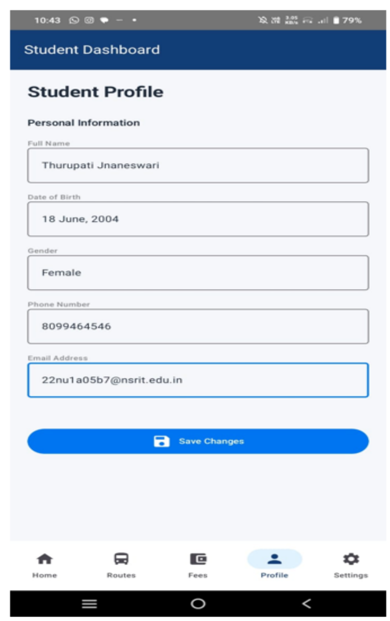
10:43
79%
Student Dashboard
Live Bus Location
Latitude : 0.000000
Longitude : 0.000000
A Bus
Tap to view live tracking
B Bus
Tap to view live tracking
C Bus
Tap to view live tracking
D Bus
Tap to view live tracking
E Bus
Tap to view live tracking
F Bus
Tap to view live tracking
Home
Routes
Fees
Profile
Settings



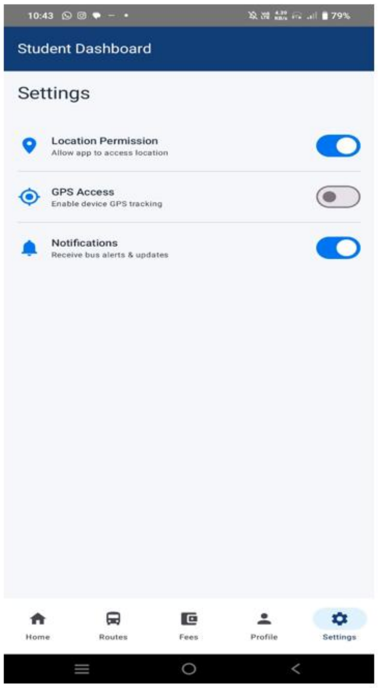
10:43
79%
Student Dashboard
Fees Payment
Student Name
Roll Number
Branch
Fees Amount (₹)
Pay Using UPI
Home
Routes
Fees
Profile
Settings



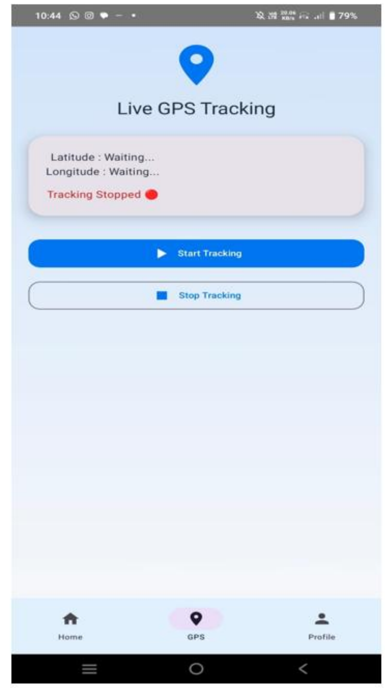
10:43
79%
Student Dashboard
Student Profile
Personal Information
Full Name
Thurupati Jnaneswari
Date of Birth
18 June, 2004
Gender
Female
Phone Number
8099464546
Email Address
student@nsrit.edu.in
Edit Profile
Home
Routes
Fees
Profile
Settings

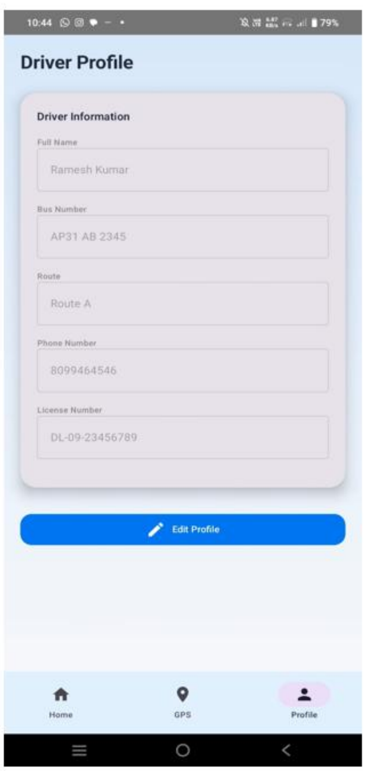


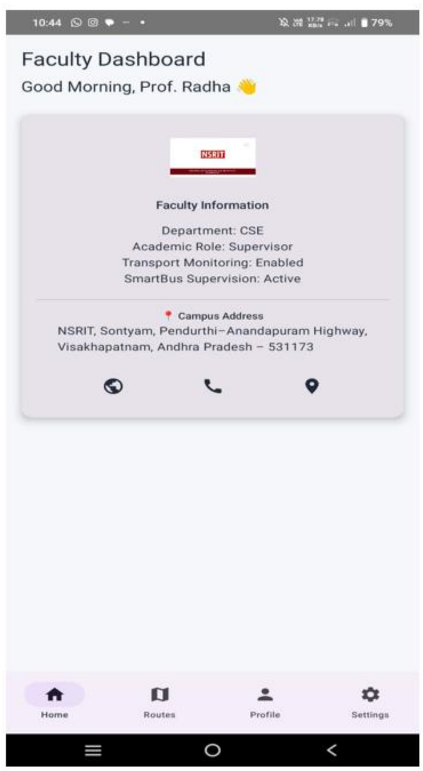
10:43
79%
Student Dashboard
Student Profile
Personal Information
Full Name
Thurupati Jnaneswari
Date of Birth
18 June, 2004
Gender
Female
Phone Number
8099464546
Email Address
22nu1a05b7@nsrit.edu.in
Save Changes
Home
Routes
Fees
Profile
Settings

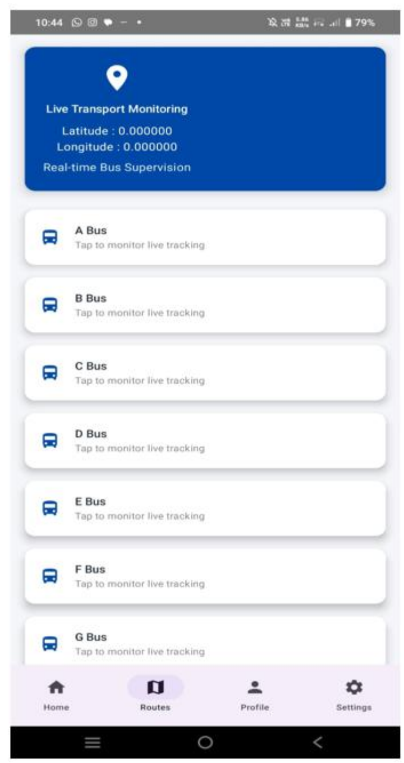

Student Dashboard
Settings
Location Permission
Allow app to access location
GPS Access
Enable device GPS tracking
Notifications
Receive bus alerts & updates
Home
Routes
Fees
Profile
Settings


Driver Dashboard
Good Morning, Ramesh
Driver Details
Bus Number : BUS-01
Route : A Route
Tracking Status : Active
About NSRIT
• Autonomous engineering institute
• Established in 2008
• UG, PG & Diploma Programs
• Strong placement support
• Smart Bus Tracking Enabled
Address
NSRIT, Sontyam, Pendurthi–Anandapuram Highway,
Visakhapatnam, Andhra Pradesh – 531173
Home
GPS
Profile


Live GPS Tracking
Latitude : Waiting...
Longitude : Waiting...
Tracking Stopped
Start Tracking
Stop Tracking
Home
GPS
Profile


Driver Profile
Driver Information
Full Name
Ramesh Kumar
Bus Number
AP31 AB 2345
Route
Route A
Phone Number
8099464546
License Number
DL-09-23456789
Edit Profile
Home
GPS
Profile


Faculty Dashboard
Good Morning, Prof. Radha
Faculty Information
Department: CSE
Academic Role: Supervisor
Transport Monitoring: Enabled
SmartBus Supervision: Active
Campus Address
NSRIT, Sontyam, Pendurthi–Anandapuram Highway,
Visakhapatnam, Andhra Pradesh – 531173
Home
Routes
Profile
Settings


Live Transport Monitoring
Latitude : 0.000000
Longitude : 0.000000
Real-time Bus Supervision
A Bus
Tap to monitor live tracking
B Bus
Tap to monitor live tracking
C Bus
Tap to monitor live tracking
D Bus
Tap to monitor live tracking
E Bus
Tap to monitor live tracking
F Bus
Tap to monitor live tracking
G Bus
Tap to monitor live tracking
Home
Routes
Profile
Settings

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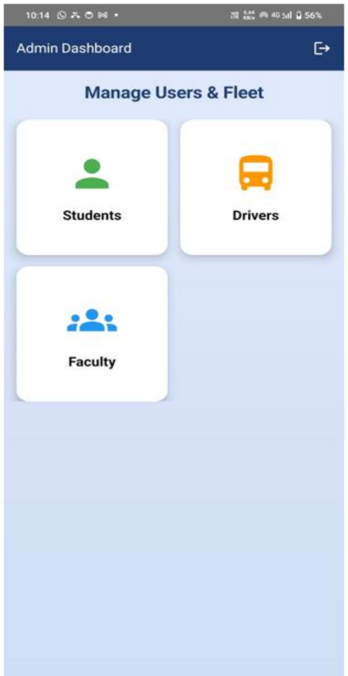

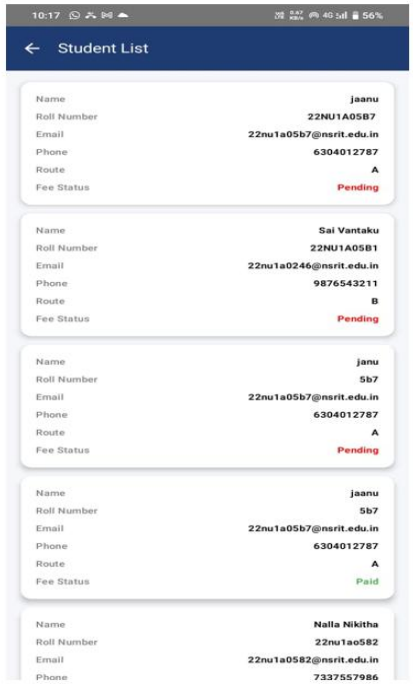

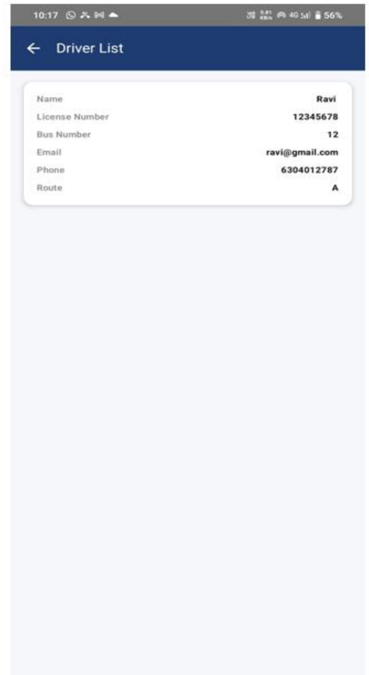

The results indicate that:

1. Location updates are reflected in real time with minimal latency.
2. ETA predictions remain accurate within an acceptable deviation range.
3. Role-based authentication successfully restricts unauthorized access.
4. Google Maps integration provides smooth and continuous bus marker updates.
5. Data synchronization remains stable under moderate internet connectivity.
6. The Admin Portal effectively manages user data, login activities, and system monitoring.
7. The digital fee payment module operates securely, with successful transaction handling and record maintenance.

Overall, the system demonstrates reliable performance, efficient real-time tracking, secure data handling, and improved user convenience. It provides a scalable and cost-effective solution for smart campus transportation management.

8. Conclusion

The NSRIT SmartBus Tracker is a scalable, cost-effective, and secure smart campus transportation solution that integrates real-time bus tracking, centralized administration, and digital fee management within a single Android-based platform. By adopting a Virtual GPS Driver Model, the system eliminates the need for dedicated GPS hardware and utilizes the driver's smartphone for accurate and continuous location transmission, significantly reducing infrastructure and maintenance costs.
The integration of Firebase services and Google Maps API enables real-time data synchronization, live bus tracking, accurate ETA calculation, and secure role-based access control. The inclusion of an Admin Portal enhances system efficiency by providing centralized monitoring, user management, and operational control. Additionally, the digital fee payment module simplifies the payment process, ensures transparency, and maintains secure transaction records.
Overall, the system improves transportation efficiency, reduces waiting time, enhances student safety, and streamlines administrative processes. It serves as a reliable and scalable solution for modern smart campus transportation management.

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Reference for implementing notification alerts related to bus delays or emergency updates (future enhancement).